

Appendix A1 - Ningaloo Marine Park Natural Values Report

North-west Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table A1.1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	
					Count	Count	Length
Aug 2019	2019-08 Ningaloo-deep stereoBRUVs	BRUV	Coastal waters, MUZ, NPZ, RUZ	132	132	123	117
May 2021	2021-05 PtCloates stereo-BRUVs	BRUV	MUZ, NPZ	25	25	25	24
May 2021	2021-05 PtCloates BOSS	BOSS	MUZ, NPZ	39	39	-	-
Aug-Sep 2021	2021-08 Pt-Cloates Flasher-BOSS	BOSS	NPZ	67	67	-	-
Aug-Sep 2021	2021-08 Pt-Cloates stereo-BRUVs	BRUV	NPZ	47	47	45	44
Sep 2021	2021-08 Pt-Cloates Squid-BOSS	BOSS	NPZ	30	30	-	-
May 2022	2022-05 PtCloates Naked-BOSS	BOSS	NPZ	74	74	-	-
May 2022	2022-05 PtCloates Squid-BOSS	BOSS	NPZ	74	65	-	-
May 2022	2022-05 PtCloates stereo-BRUVs	BRUV	NPZ	64	64	62	62
May 2024	DR2024-02 CSIRO Ningaloo stereoBRUVs	BRUV	MUZ, NPZ, RUZ	71	65	63	55

Benthic ecosystem extent

Observations

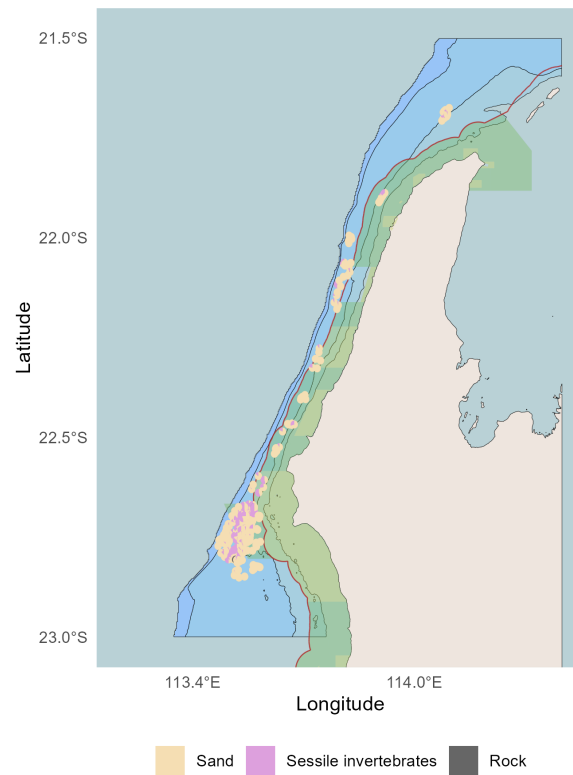


Figure A1.1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

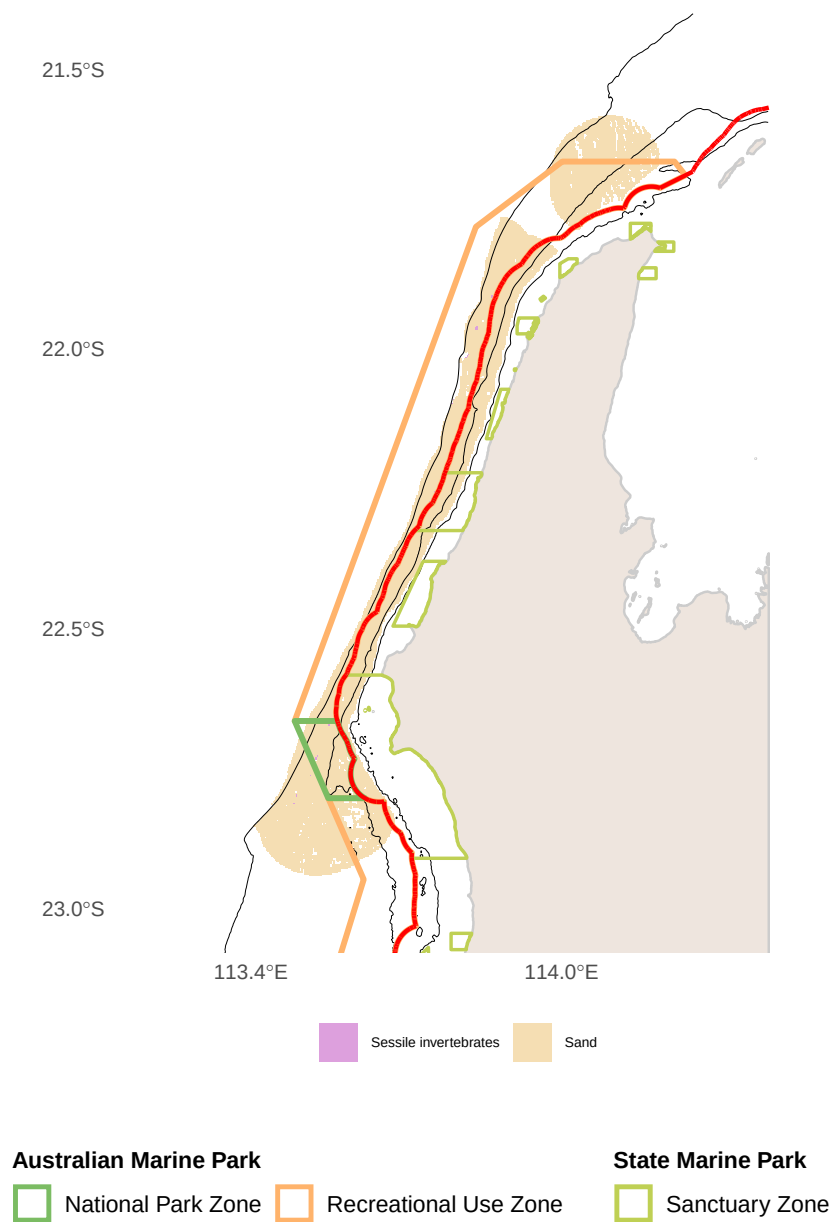


Figure A1.2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

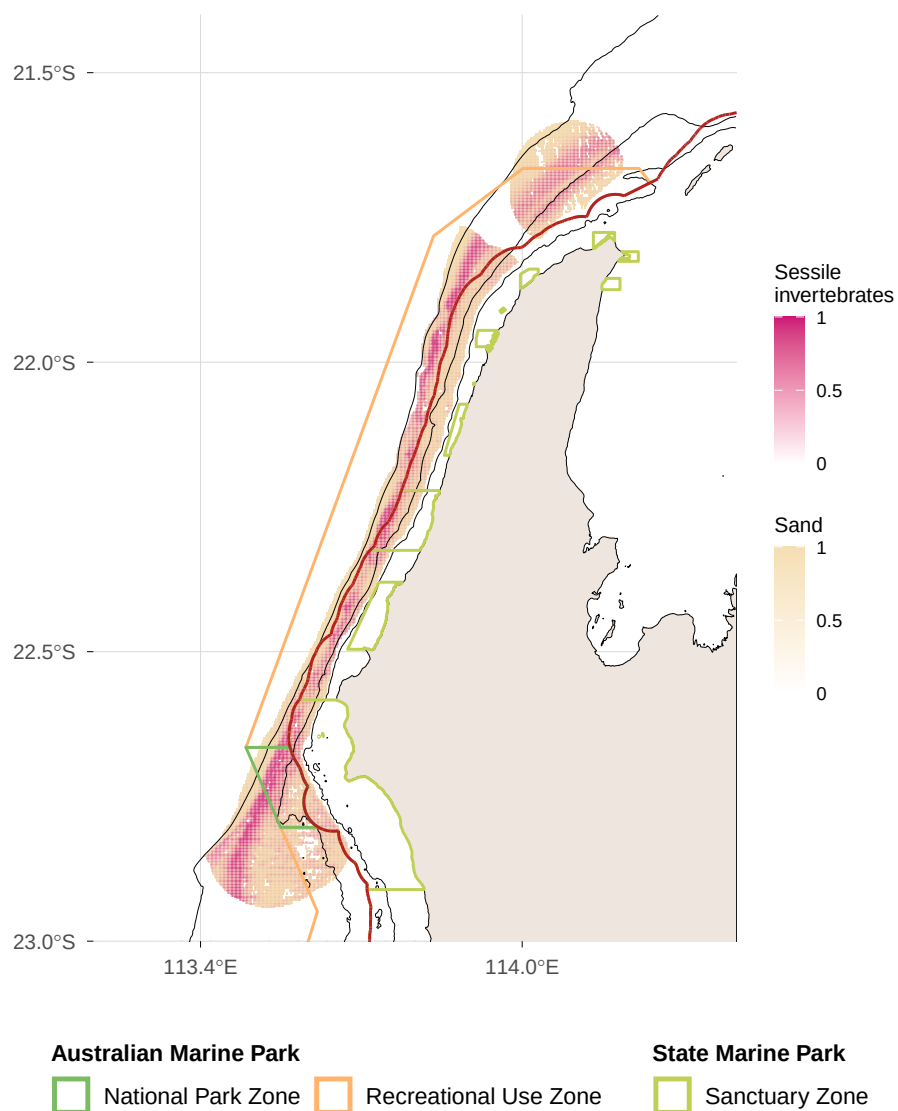


Figure A1.3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

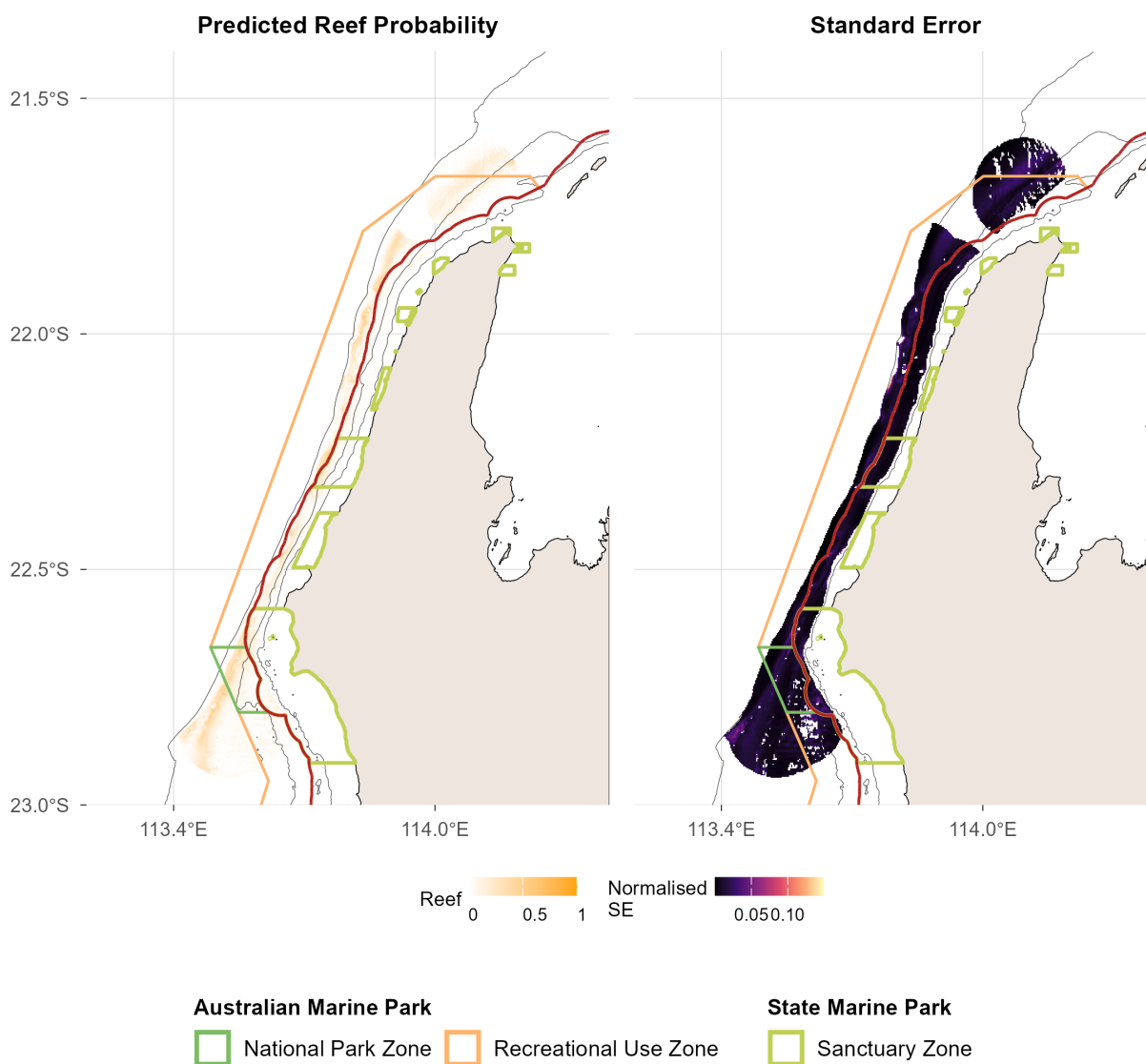


Figure A1.4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Sand

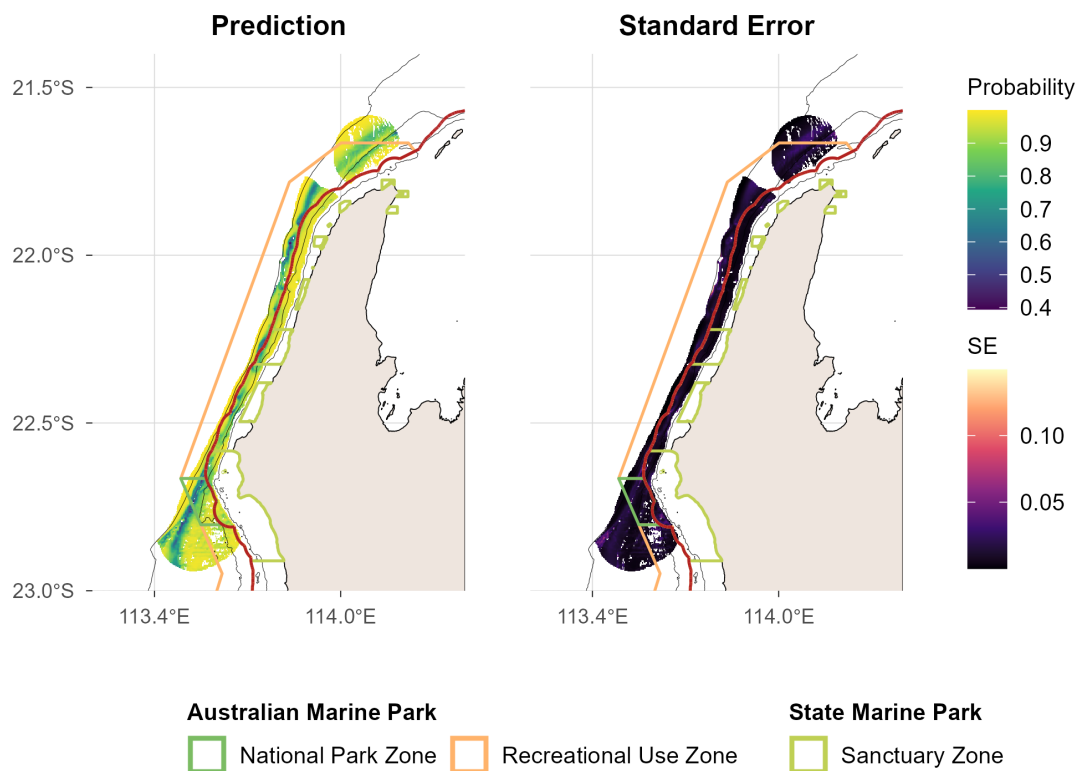


Figure A1.5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

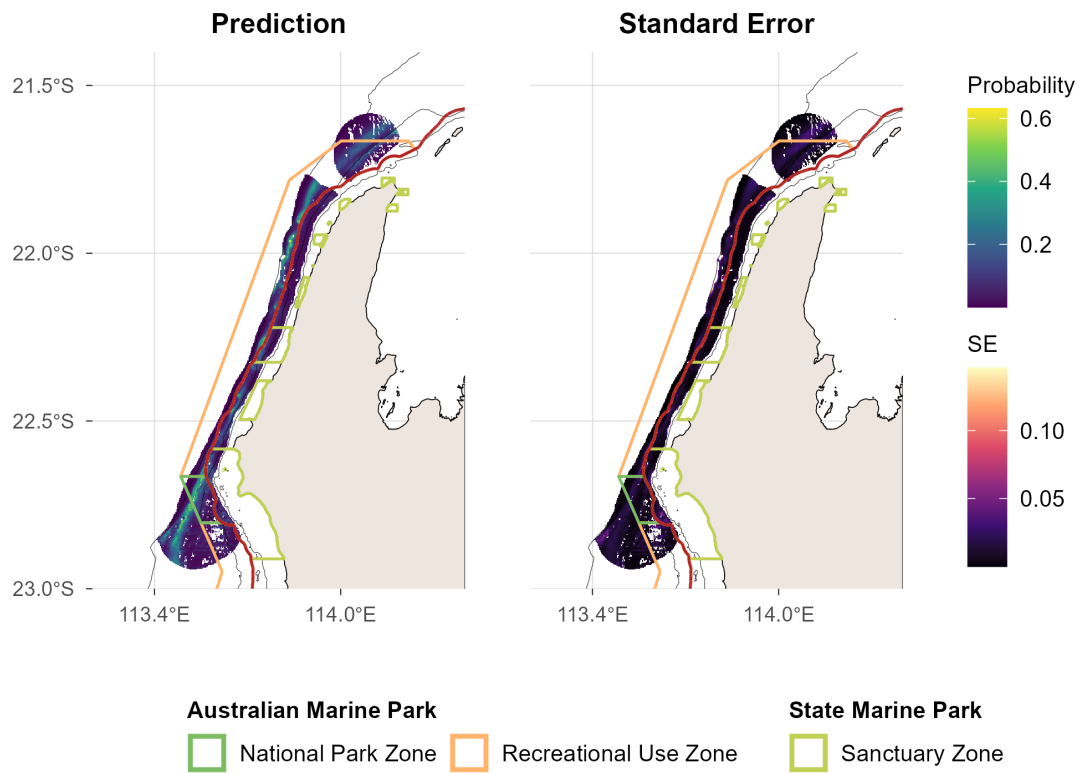


Figure A1.6.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

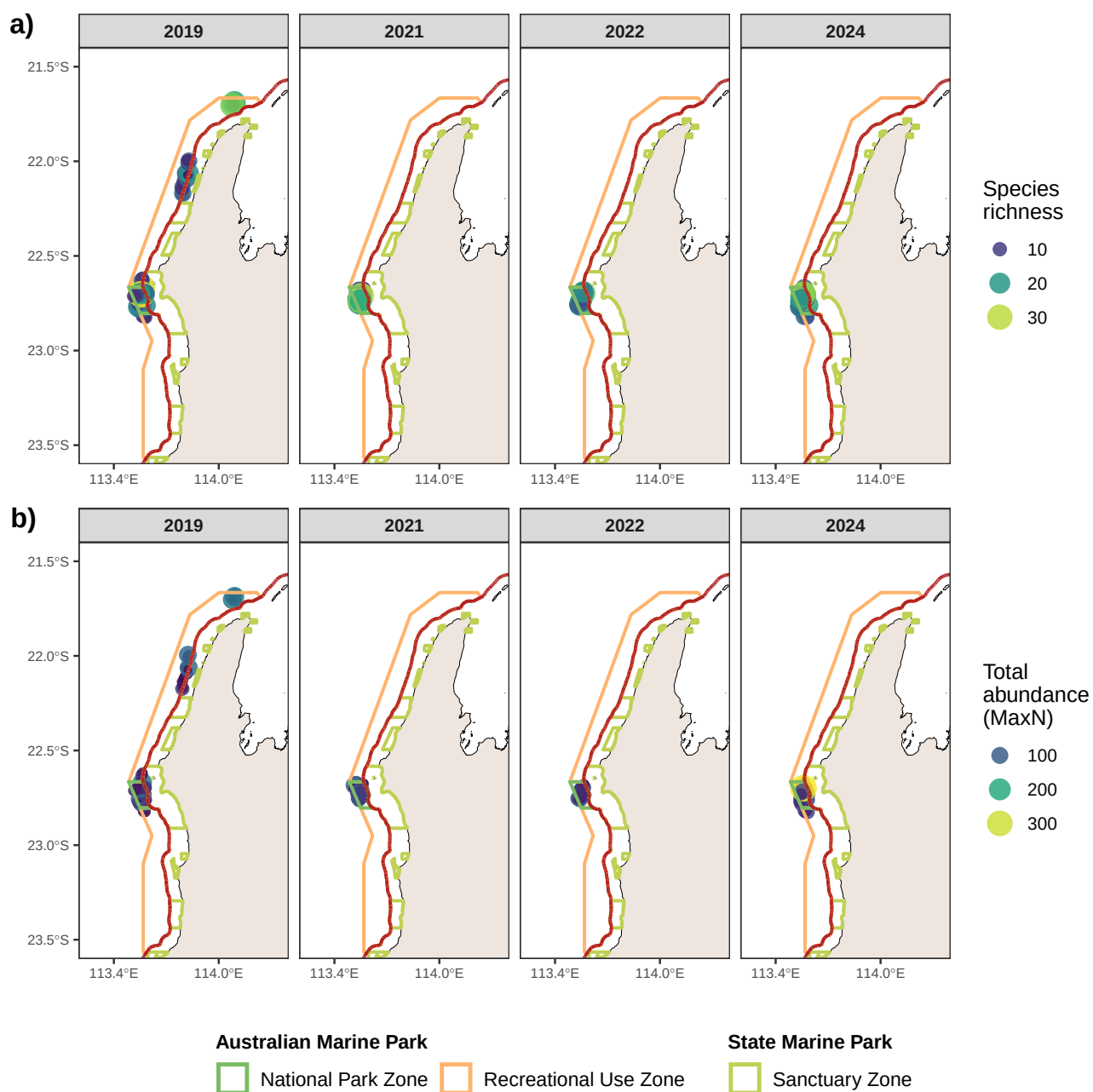


Figure A1.7: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

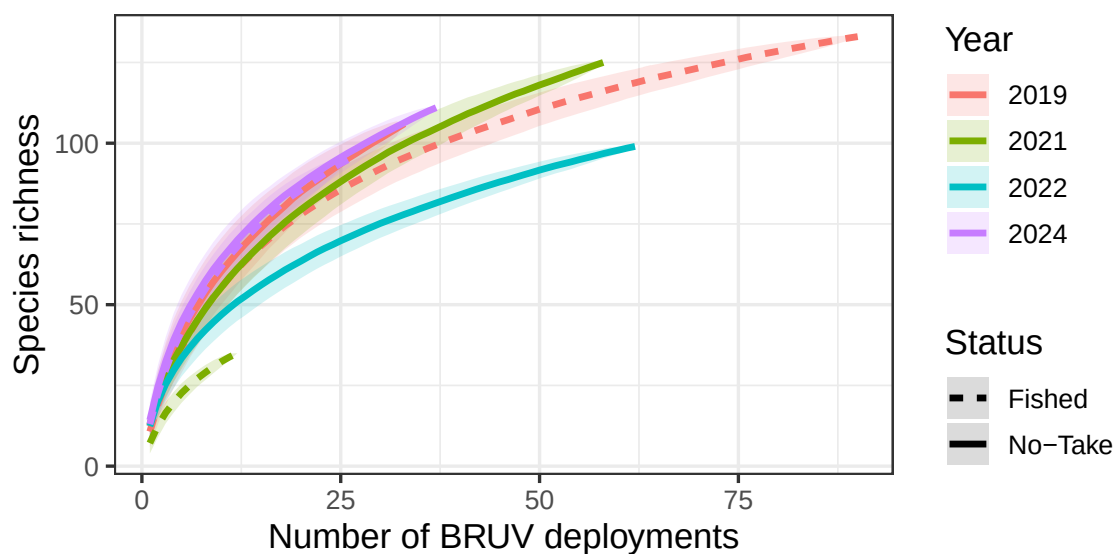


Figure A1.8.1: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth).

Species richness

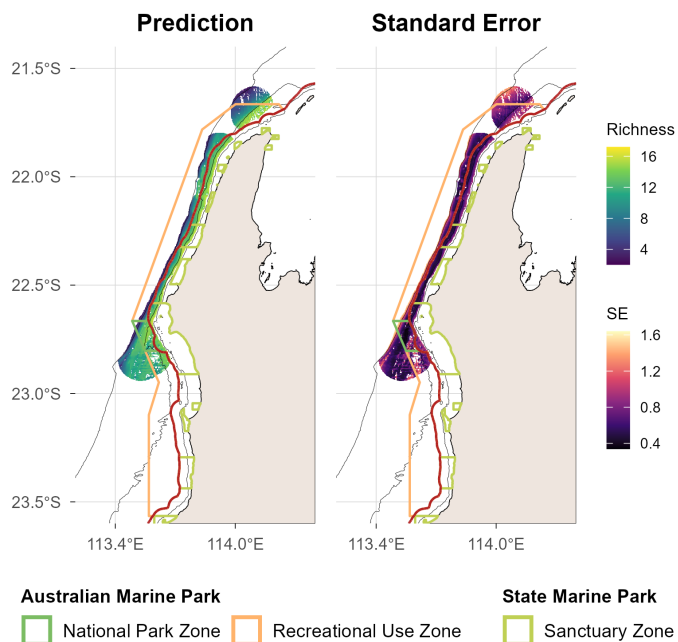


Figure A1.8.2: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

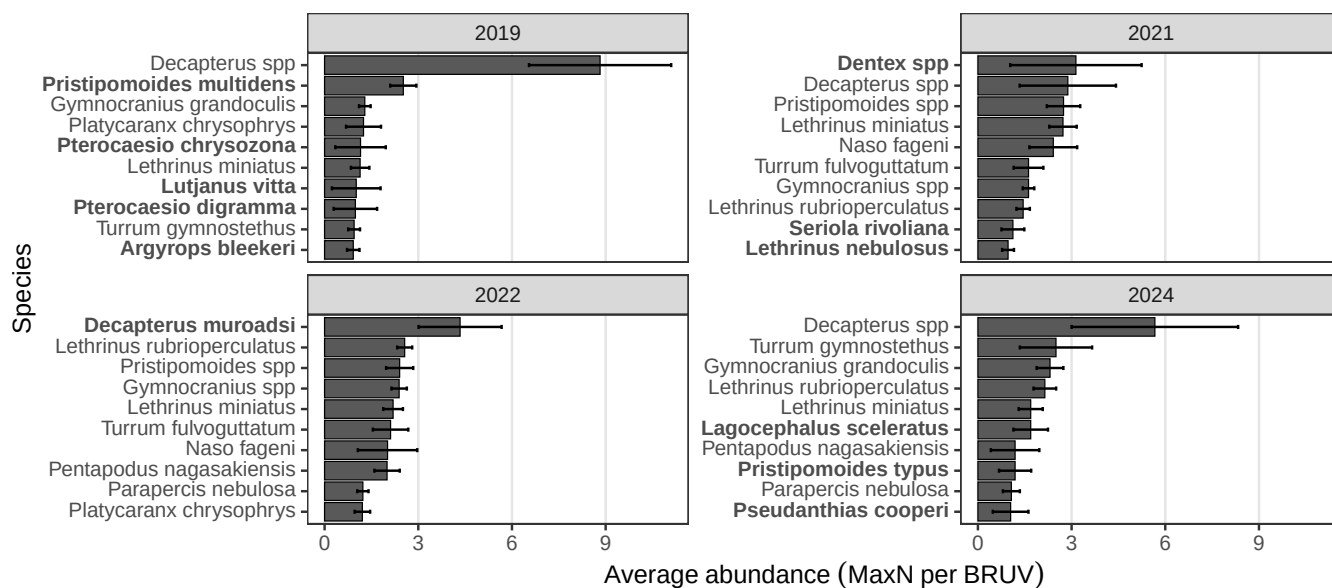


Figure A1.9.1: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth).

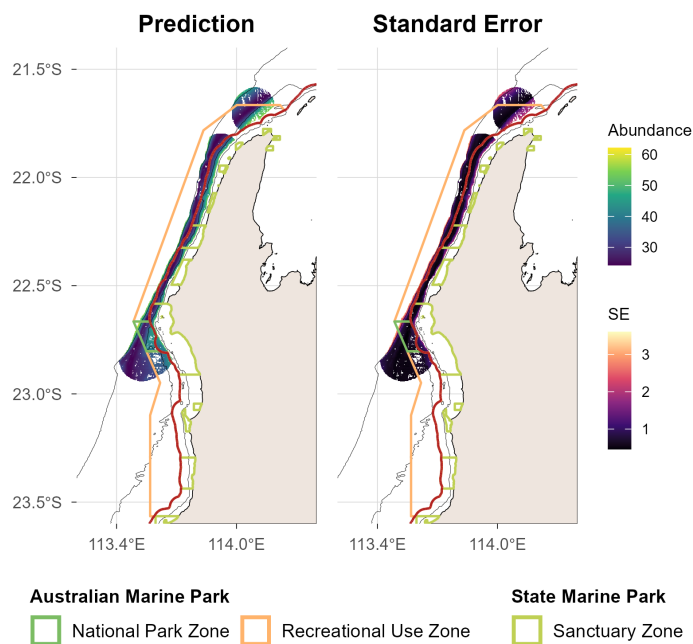


Figure A1.9.2: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

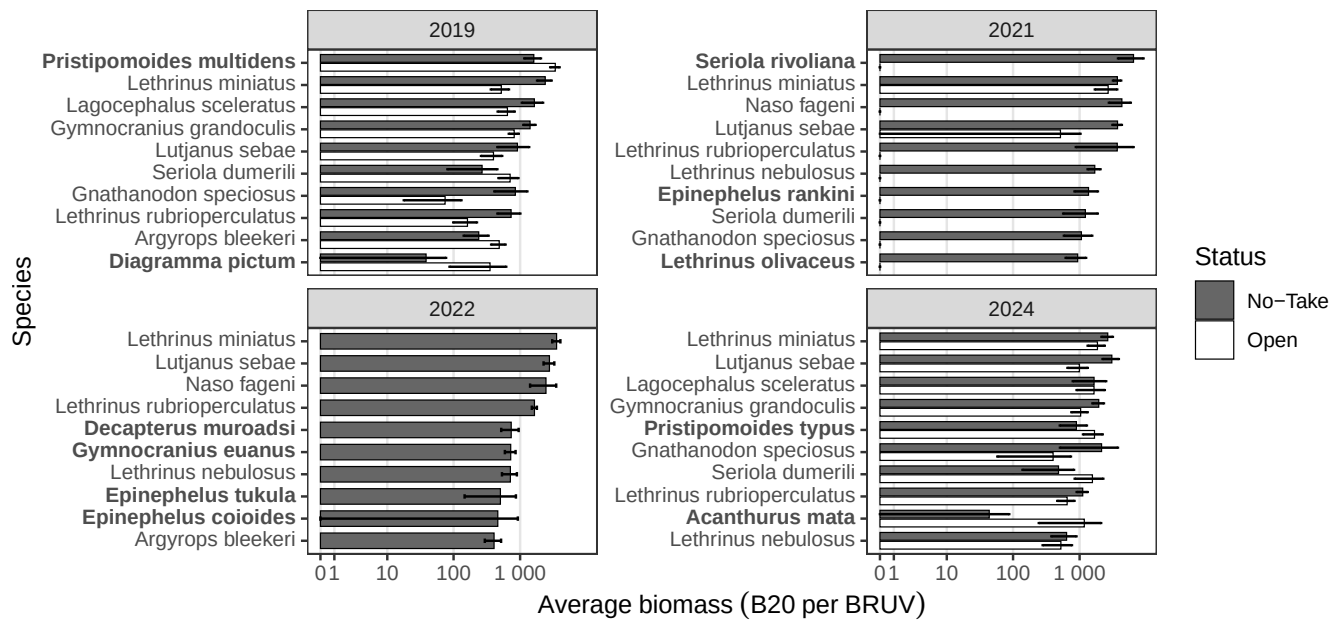


Figure A1.10.1: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

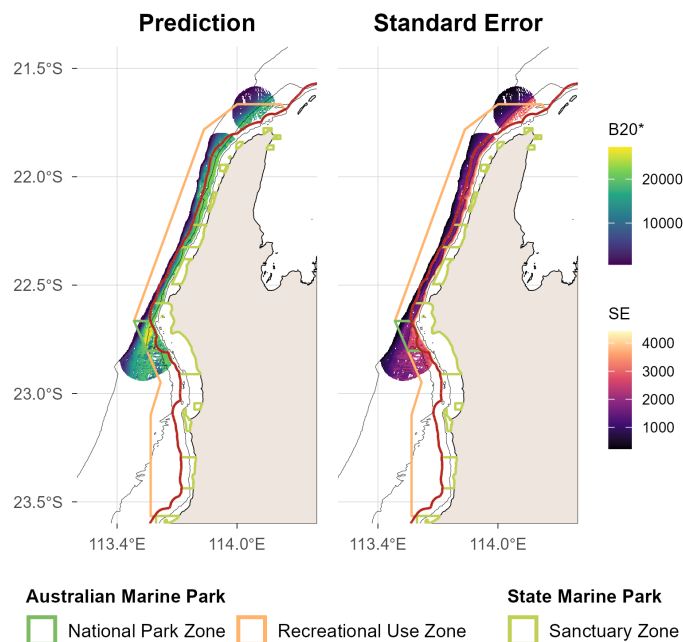


Figure A1.10.2: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

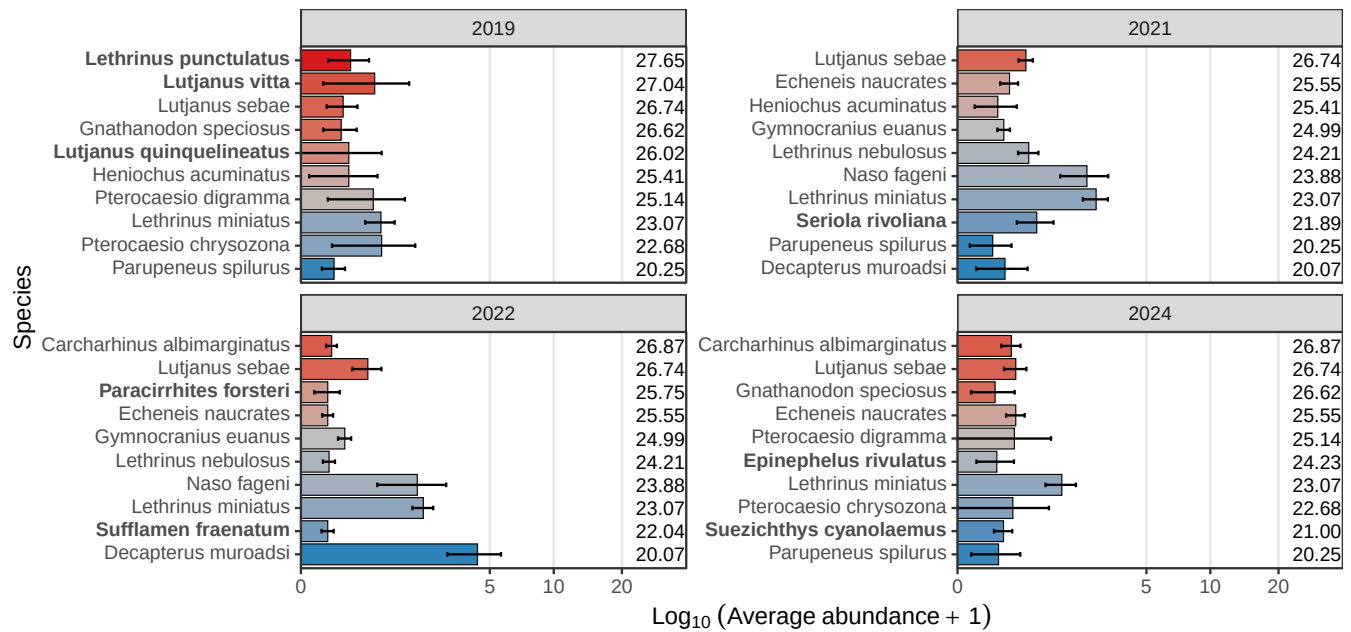


Figure A1.11.1: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. Ningaloo Marine Park (Commonwealth).

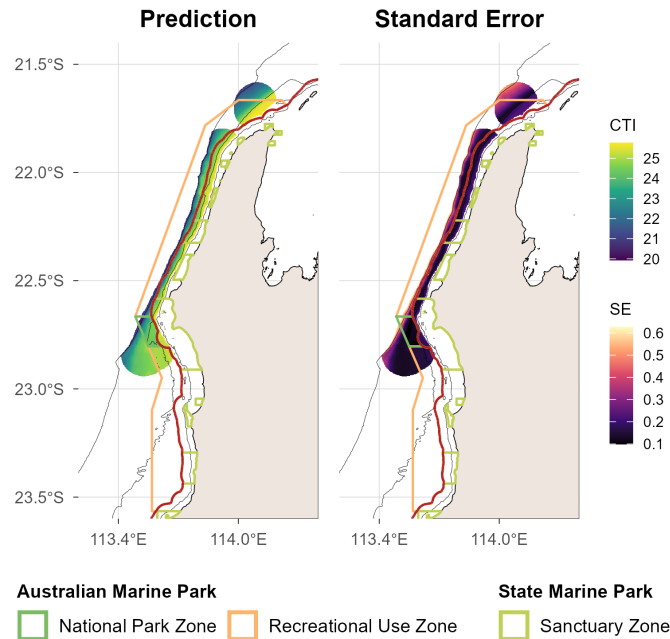


Figure A1.11.2: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Threatened species

Table A1.2: Threatened species recorded in stereo-BRUV imagery across all survey years, showing IUCN Red List category, the number of samples each species was recorded in, and the total abundance (MaxN) summed across those samples. *This species is also listed as ‘Conservation dependent’ on the EPBC Act List of Threatened Fauna. ^This species is also listed as ‘Vulnerable’ on the EPBC Act List of Threatened Fauna.

Species	Common name	Family	IUCN category	Samples	Total MaxN	Years recorded
<i>Carcharhinus albimarginatus</i>	Silvertip Shark	Carcharhinidae	Vulnerable	62	80	2019, 2021, 2022, 2024
<i>Carcharhinus plumbeus</i>	Sandbar Shark	Carcharhinidae	Endangered	43	45	2019, 2021, 2022, 2024
<i>Loxodon macrorhinus</i>	Sliteye Shark	Carcharhinidae	Near Threatened	20	26	2019, 2021, 2022, 2024
<i>Galeocerdo cuvier</i>	Tiger Shark	Carcharhinidae	Near Threatened	12	12	2019, 2021, 2022, 2024
<i>Sphyrna mokarran</i> *	Great Hammerhead	Sphyrnidae	Critically Endangered	12	12	2019, 2022, 2024
<i>Rhynchobatus australiae</i>	Whitespotted Guitarfish	Rhinidae	Critically Endangered	6	6	2019, 2021, 2022, 2024
<i>Sphyrna lewini</i> *	Scalloped Hammerhead	Sphyrnidae	Critically Endangered	6	8	2019, 2021
<i>Scomberomorus commerson</i>	Spanish Mackerel	Scombridae	Near Threatened	4	4	2019, 2021, 2022
<i>Carcharhinus limbatus</i>	Common Blacktip Shark	Carcharhinidae	Vulnerable	3	3	2019, 2021, 2024
<i>Rhizoprionodon acutus</i>	Milk Shark	Carcharhinidae	Vulnerable	3	4	2019, 2024
<i>Taeniurops meyeri</i>	Blotched Fantail Ray	Dasyatidae	Vulnerable	3	3	2021, 2024
<i>Carcharhinus macroti</i>	Hardnose Shark	Carcharhinidae	Near Threatened	2	3	2021
<i>Carcharhinus amblyrhynchos</i>	Grey Reef Shark	Carcharhinidae	Endangered	1	1	2021
<i>Carcharias taurus</i>	Greynurse Shark	Carchariidae	Critically Endangered	1	1	2021
<i>Nemipterus virgatus</i>	Golden Threadfin Bream	Nemipteridae	Vulnerable	1	1	2022
<i>Pateobatis fai</i>	Pink Whipray	Dasyatidae	Vulnerable	1	3	2024
<i>Prionace glauca</i>	Blue Shark	Carcharhinidae	Near Threatened	1	1	2019
<i>Stegostoma tigrinum</i>	Zebra Shark	Stegostomatidae	Endangered	1	1	2022
<i>Triaenodon obesus</i>	Whitetip Reef Shark	Carcharhinidae	Vulnerable	1	1	2019

Threatened shark and ray species

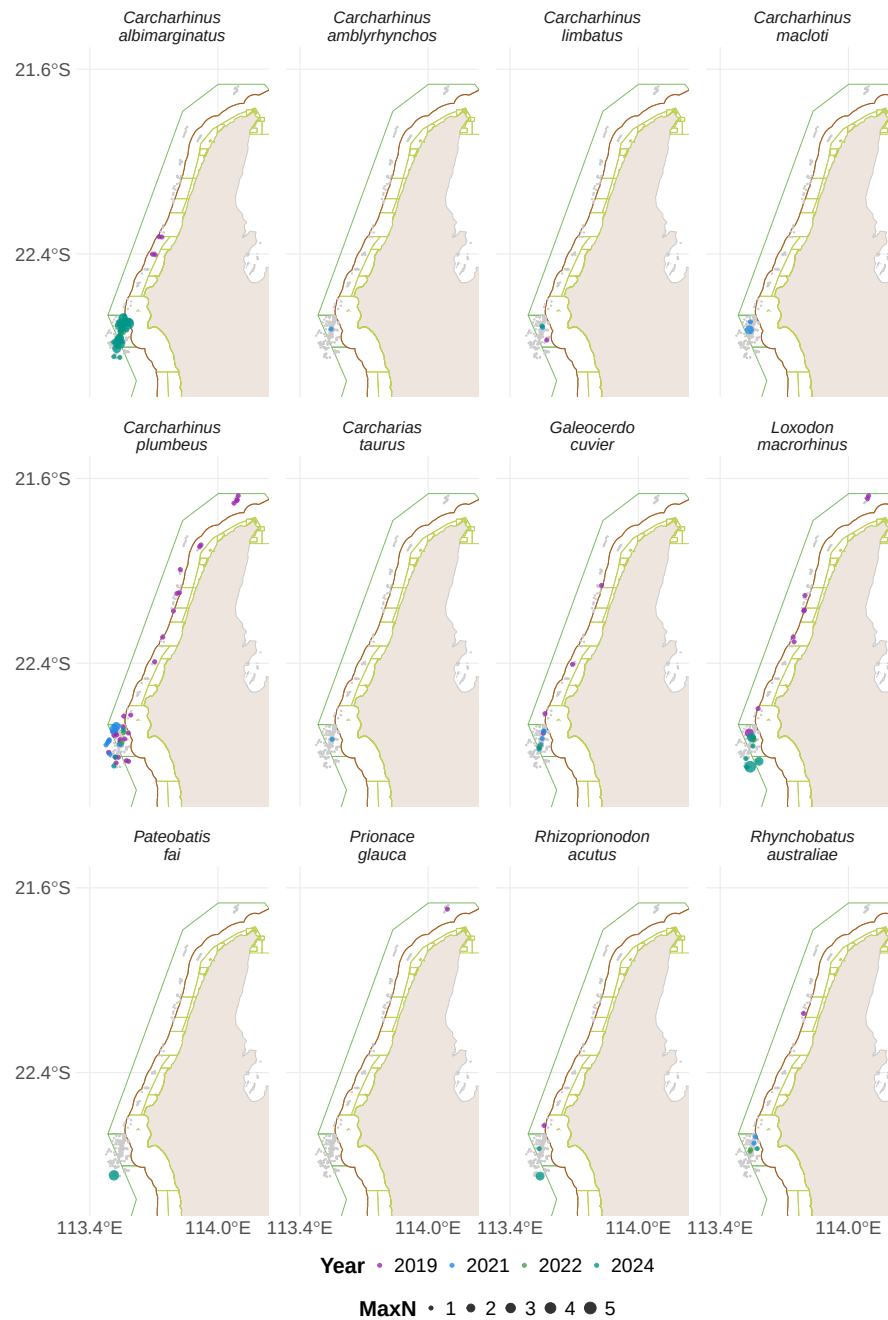


Figure A1.12.1: Occurrence of threatened, IUCN Red List category, sharks and rays in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment. Commonwealth and coastal marine park boundaries are shown. The red line delimits coastal and Commonwealth waters.

Other shark and ray species

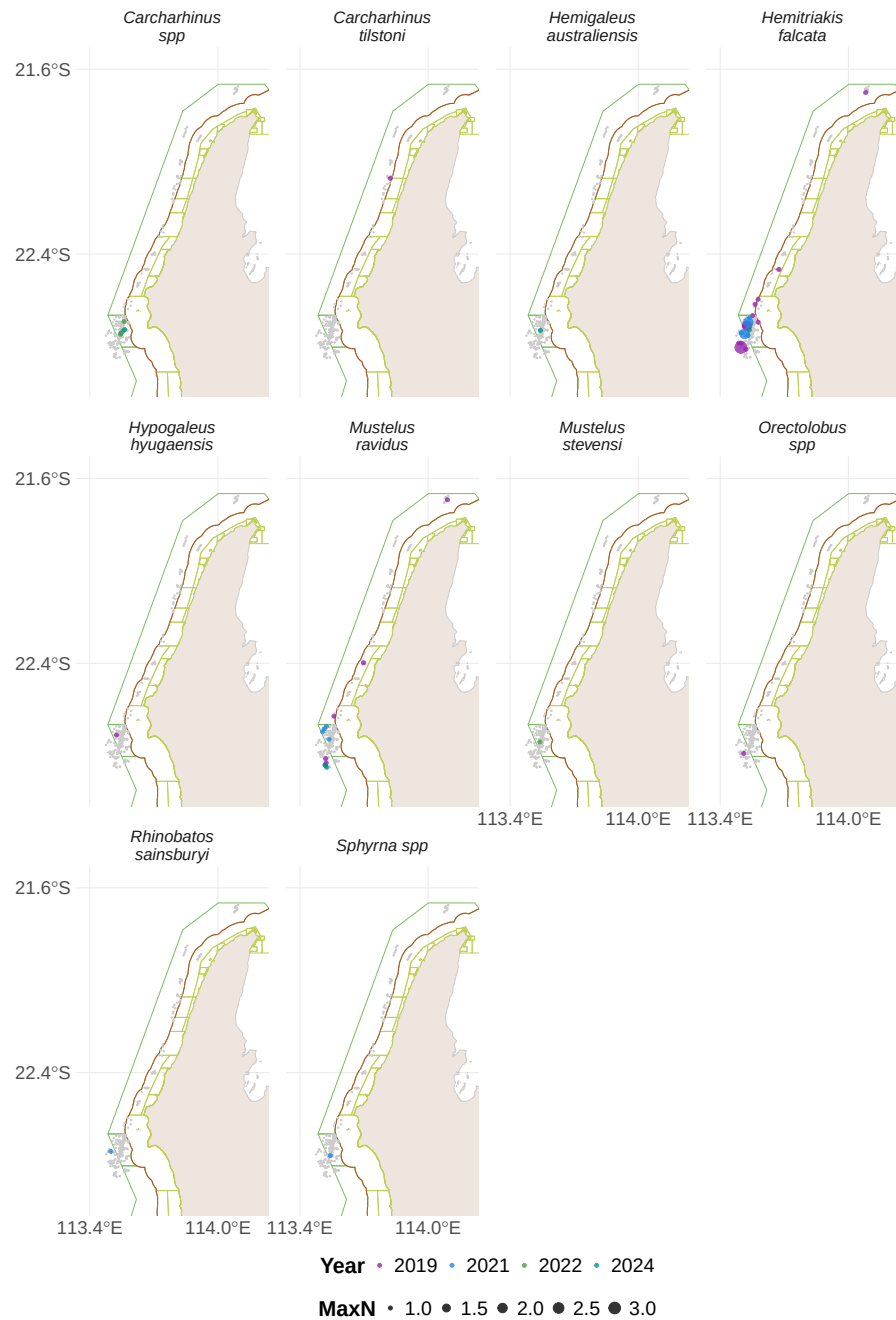


Figure A1.12.2: Occurrence of shark and ray species that are not threatened in stereo-BRUV imagery. Points are scaled by abundance (MaxN) and coloured by survey year; grey points show every stereo-BRUV deployment, so a panel with no coloured points is a genuine absence rather than an absence of sampling. Commonwealth and state marine park boundaries are shown. The red line delimits state and commonwealth waters.